# Python Engineering - Engineering & Math Theory

## Class 12 - Engineering Toolkit: essential vector algebra

Mathematical theory behind EX20: magnitude, normalization, dot product, angle between vectors, projection, and validation of rotation matrices.

## 1. Vector norm or magnitude

For v=[x,y], the Euclidean norm is:

$$||v|| = \sqrt{x^2 + y^2}$$

Manual example: v=[3,4] -> ||v||=sqrt(9+16)=5. In code we use np.linalg.norm(v).

## 2. Unit vector

A unit vector preserves direction but has magnitude 1:

$$\hat{v} = v / ||v||$$

For [3,4], v_hat=[0.6,0.8]. The zero vector cannot be normalized because it would require division by zero.

## 3. Dot product

For vectors a and b:

$$a \cdot b = a_x b_x + a_y b_y$$

Geometrically:

$$a \cdot b = ||a|| \, ||b|| \cos(\theta)$$

If the dot product is zero and both vectors are non-zero, they are perpendicular.

## 4. Angle between vectors

Solving the dot-product relation for theta:

$$\theta = \arccos\left( (a \cdot b) / (||a|| \, ||b||) \right)$$

Example: a=[3,4], b=[1,0]. a.b=3, ||a||=5, ||b||=1, so cos(theta)=0.6 and theta≈53.1301 deg.

## 5. Why do we need np.clip()?

Mathematically cosine lies in [-1,1], but floating-point rounding can produce 1.0000000000000002. arccos is not real-valued there even though the intended value is 1.

Therefore we clip cos_theta to [-1,1] before np.arccos().

## 6. Vector projection

Projection of a onto direction b:

$$\text{proj}_b(a) = (a \cdot b / b \cdot b) \, b$$

Example: a=[3,4], b=[1,0] -> proj_b(a)=[3,0]. Physically this can represent the component of force, velocity, or acceleration along an axis.

## 7. Validating a rotation matrix

A proper rotation matrix should satisfy:

$$R^T R = I$$

$$\det(R) = +1$$

The first condition checks orthogonality. The second excludes reflections: a reflection may also be orthogonal but has determinant -1.

## 8. Why is R^T R = I not enough?

The matrix [[1,0],[0,-1]] satisfies R^T R=I but reflects the y axis rather than performing a proper rotation. Its determinant is -1. Therefore check_rotation_matrix() checks both properties.

## 9. Direct connection to EX20

| Toolkit function | Mathematical theory |
|---|---|
| vector_norm() | Euclidean norm ||v|| |
| unit_vector() | Normalization v/||v|| |
| angle_between_vectors() | Dot product and cos(theta) |
| project_vector() | Vector projection |
| rotation_matrix_2d() | Orthogonal linear transformation |
| check_rotation_matrix() | R^T R=I and det(R)=1 |

## 10. Why build a toolkit?

Engineering projects repeatedly use these operations. Encapsulating them avoids duplicated formulas, reduces errors, and allows each tool to be validated independently before reuse in dynamics, navigation, sensors, astrodynamics, or numerical methods.